



Lithium isotope stratigraphy and Ordovician weathering

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ABSTRACT

Understanding the role of silicate weathering in controlling the long-term carbon cycle is essential in exploring multi-million-year scale climate variabilities and links to tectonics and changes in biodiversity. The lithium isotopic composition of carbonate rocks ($\delta^7\text{Li}$) is a non-traditional proxy that offers a more comprehensive understanding of silicate weathering dynamics in the geologic past in combination with traditional weathering proxies such as strontium isotopes ($^{87}\text{Sr}/^{86}\text{Sr}$). However, questions related to the effects of diagenesis and the relationship between changes in $\delta^7\text{Li}$ and CO_2 consumption during silicate rock weathering hamper the use of this novel proxy to its full potential. Here we present measurements of bulk carbonate $\delta^7\text{Li}$ that span the Ordovician Period from six sections across the Laurentian paleocontinent. Using a rigorous textural and geochemical screening, we produce the first high-resolution seawater $\delta^7\text{Li}$ ($\delta^7\text{Li}_{\text{sw}}$) curve that spans the Tremadocian to Sandbian of the Ordovician Period (~487 – 453 Ma). The $\delta^7\text{Li}_{\text{sw}}$ curve produced in this study exhibits a ~4‰ negative shift during the Early to Middle Ordovician. Using comparisons to existing weathering proxies and a numerical modeling approach, we suggest a progressive change toward a low-intensity, congruent weathering regime was a likely driver for the observed trend. This finding highlights potential links between island arc migration toward the tropics, global increase in exposed landmass, possible terrestrialization by non-vascular land plants and changes in silicate weathering.

1. Introduction

Silicate weathering plays a major role in the long-term carbon cycle and the habitability of our planet (Kasting, 2019). Its role as a major carbon sink and sensitivity to temperature makes it important to characterize changes in silicate weathering in both the present-day and geologic past (Isson et al., 2020). Consequently, there has been growing interest in developing carbonate-based geochemical proxies to reconstruct changes in silicate weathering and its impact on global climate. One promising approach is the use of lithium isotopes. The lithium isotopic composition ($\delta^7\text{Li}$) is thought to be mainly affected by silicate weathering (and its reverse process; i.e., reverse weathering), while not being affected by the type of rock being weathered or fractionation by biologic processes (Dellinger et al., 2015; Pogge von Strandmann et al., 2019), which are two factors that limit the usage of more traditional radiogenic isotope weathering proxies (e.g., $^{87}\text{Sr}/^{86}\text{Sr}$, ϵ_{Nd}).

The $\delta^7\text{Li}$ value provides constraints on silicate weathering independent of lithology because fractionation of riverine $\delta^7\text{Li}$ is mainly controlled by the relative dominance of primary mineral dissolution versus clay mineral formation. Primary mineral dissolution does not fractionate Li isotopes whereas clay mineral formation preferentially incorporates ^6Li and drives riverine $\delta^7\text{Li}$ to higher values. Observations from various drainage basins show that regions with high physical erosion rates and short water residence times tend to have high Li flux and low riverine $\delta^7\text{Li}$ as a result of congruent weathering, whereas regions with flatter topography, low physical erosion rates, and long water residence times are characterized by low Li flux and high riverine $\delta^7\text{Li}$ (e.g., Dellinger et al., 2015; Pogge von Strandmann and Henderson, 2015). These observations indicate that riverine $\delta^7\text{Li}$ is tightly linked to silicate weathering intensity, which is defined as the ratio of the chemical weathering rate (W) to the denudation rate (D), where denudation rate is the sum of chemical weathering rate and physical erosion

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rate (Dellinger et al., 2015; Pogge von Strandmann et al., 2021). This implies that weathering intensity (WI) is practically a measure of the relative significance between chemical weathering (W) and physical erosion (E). Mountainous regions with high erosion rates, hence low weathering intensity, are dominated by congruent weathering processes (low $\delta^7\text{Li}$ and high Li flux), whereas flatter region such as floodplains with intense chemical weathering and low erosion rates, hence moderate weathering intensity, are dominated by incongruent weathering (high $\delta^7\text{Li}$ and lower Li flux). A high weathering intensity regime is characterized by supply-limited weathering (i.e., limited supply of fresh primary rocks to dissolve) and hence is dominated by leaching of secondary clays, which results in low $\delta^7\text{Li}$ and very low Li flux (Dellinger et al., 2015). Due to the low Li flux, a high weathering intensity regime is thought to have a less direct influence on the $\delta^7\text{Li}$ of seawater. Furthermore, studies have suggested that weathering intensity is tightly linked to the efficiency of CO_2 removal via silicate weathering (e.g., Caves Rugenstein et al., 2019; Pogge von Strandmann et al., 2021). As such, a number of studies use this framework to interpret stratigraphic changes in seawater $\delta^7\text{Li}$ ($\delta^7\text{Li}_{\text{sw}}$) values as a signal of changes in the efficiency of continental silicate weathering in consuming CO_2 and modulating climate in the geologic past (Pogge von Strandmann et al., 2021, 2017; Sproson et al., 2022; Sun et al., 2018; Wang et al., 2021).

Despite the promise of using Li isotopes in marine carbonates as a proxy for global silicate weathering intensity, major challenges remain in how the data are interpreted. Some studies have suggested that CO_2 consumption via silicate weathering can be interfered with by oxidative weathering of sulfides and organic carbon (Georg et al., 2013; Sproson et al., 2022). This process, however, does not seem to have a straightforward relationship with atmospheric CO_2 and variabilities in riverine $\delta^7\text{Li}$ (Ramos et al., 2022). Therefore, comparing $\delta^7\text{Li}$ data to previously published weathering proxies during well-constrained changes in global silicate weathering can elucidate a more nuanced understanding of the links between weathering and the Li cycle.

The impact of diagenetic processes in altering the primary signals of $\delta^7\text{Li}$ in carbonate rocks, the most common sample material used in deep time studies, remains an important question. For instance, a study in the modern Bahamas suggests that Li isotope values in carbonate rocks are sensitive to meteoric and early marine diagenesis (Dellinger et al., 2020; Murphy et al., 2022), such that reconstructing the primary seawater $\delta^7\text{Li}$ requires a numerical modeling approach based on a suite of samples with varying diagenetic conditions (Murphy et al., 2022). Studies in deep time have relied mainly on elemental and textural evidence to constrain diagenesis (Cao et al., 2022; Kalderon-Asael et al., 2021; Pogge von Strandmann et al., 2017; Sun et al., 2018; Wang et al., 2021). Even though diagenesis seems to affect modern carbonate samples, a recent study shows that Neogene dolostones are able to preserve the seawater lithium isotopic composition (Liu et al., 2023). However, although the $\delta^7\text{Li}_{\text{sw}}$ value should be a global signal, reflecting the long residence time of lithium in the oceans (~ 1.2 million years; Misra and Froelich, 2012), there are relatively few lithium isotope studies that have compared regional and global variabilities on a multi-million year timescale, which are ultimately required to further constrain the robustness of carbonate $\delta^7\text{Li}$ ($\delta^7\text{Li}_{\text{carb}}$) as a proxy for $\delta^7\text{Li}_{\text{sw}}$.

The uncertainties in interpreting changes in $\delta^7\text{Li}_{\text{sw}}$ in the context of silicate weathering and challenges in using carbonate rocks to reconstruct ancient $\delta^7\text{Li}_{\text{sw}}$ highlight the need for studies that explore the $\delta^7\text{Li}_{\text{carb}}$ proxy during periods of significant changes in global silicate weathering and regional comparisons from chronostratigraphically well-constrained sections. Here we analyze stratigraphic variations in $\delta^7\text{Li}$ of carbonate rocks that span the Lower to Upper Ordovician interval from six chronostratigraphically well-constrained sections across the Laurentian paleocontinent. The Ordovician is an ideal case study to explore the relationships between $\delta^7\text{Li}_{\text{sw}}$ and silicate weathering as this period is characterized by significant changes in global silicate weathering using other independent methods (Avila et al., 2022; Conwell

et al., 2022; Pogge von Strandmann et al., 2017; Swanson-Hysell and Macdonald, 2017).

Here lithium isotope measurements are paired with major and trace element ratios to screen for detrital contamination (Li/Ca, Al/Ca, Rb/Sr, and Rb/Ca) and diagenetically altered samples (Mg/Ca, Fe/Ca, Mn/Ca, Mn/Sr, Sr/Ca). We use the least altered samples to construct an Ordovician $\delta^7\text{Li}_{\text{sw}}$ curve utilizing a bootstrapped LOESS method, and compare to previously published proxy evidence for changes in silicate weathering (Conwell et al., 2022; Swanson-Hysell and Macdonald, 2017) and paleotemperature trends (Avila et al., 2022) during the Ordovician. The result of Ordovician $\delta^7\text{Li}_{\text{sw}}$ curve reconstruction done in this study exhibits a 4‰ negative shift from 10‰ at the base of Tremadocian Stage to 6‰ at the lower portion of the Floian Stage. This declining trend is broadly coeval with the observed cooling trend during the Early Ordovician and changes in $^{87}\text{Sr}/^{86}\text{Sr}$ toward more non-radiogenic values. A simple stochastic model developed in this study indicates that a shift toward more congruent weathering is a likely driver of the observed declining trend in $\delta^7\text{Li}_{\text{sw}}$. This result suggests that changes in the Li cycle during the Ordovician was largely due to changes in continental silicate weathering instead of changes in silicon (Si) cycles as previously suggested (Kalderon-Asael et al., 2021). Additionally, the shift toward more congruent weathering regime may have contributed to the observed global cooling trend during the Early to Middle Ordovician (Fig. 1).

The Ordovician (487 – 443 Ma) records an estimated 15 °C cooling (Fig. 1) that culminated in the Hirnantian Glaciation (e.g., Grossman and Joachimski, 2020). Several hypotheses have been put forward to explain the cause of the Ordovician cooling including: reduced volcanic outgassing (McKenzie et al., 2016) and increased silicate weathering rates (Conwell et al., 2022; Nardin et al., 2011; Swanson-Hysell and Macdonald, 2017; Young et al., 2009). While constraining changes in volcanic outgassing requires a multitude of assumptions, recent advances in “full-plate” tectonic models suggest that the early Paleozoic is characterized by a reduction in volcanic activity (Marcilly et al., 2021). Alternatively, a number of studies have proposed evidence that supports increased silicate weathering, which includes: high-resolution geochemical weathering proxy dataset ($^{87}\text{Sr}/^{86}\text{Sr}$ and ϵ_{Nd}), numerical modeling, and refined paleogeographic reconstructions (Fig. 1) (Conwell et al., 2022; Nardin et al., 2011; Swanson-Hysell and Macdonald, 2017).

While enhanced silicate weathering could potentially explain the cooling trend observed in the Middle Ordovician (Conwell et al., 2022; Nardin et al., 2011; Swanson-Hysell and Macdonald, 2017), the cause of the broader cooling trend characterizing the Early Ordovician climate remains ambiguous. A modeling study presented in Conwell et al. (2022) for instance, relies on a forced decrease in volcanic outgassing flux to reproduce the observed cooling during the Early Ordovician. Therefore, constraining a potential role for silicate weathering in causing the Early Ordovician cooling using $\delta^7\text{Li}$ values can potentially shed light on the Ordovician paleoclimate evolution.

2. Geologic background and stratigraphy

We analyzed carbonate rocks from six sections across the Laurentian paleocontinent, with four sections located in the Great Basin region and two in the Appalachian Basin (Fig. 2). These sections represent carbonate rocks from a passive margin setting (Great Basin) and foreland basin (Appalachian Basin). A recent paleogeographic reconstruction suggests that these sections were located in the tropics during the Early Ordovician (Swanson-Hysell and Macdonald, 2017). The studied sections span the Tremadocian to Sandbian stages (Early to Late Ordovician), and have relatively robust chronostratigraphic constraints based on biostratigraphy (Harris et al., 1979; Miller et al., 2003) and chemostratigraphy (Adiatma et al., 2019; Diamond et al., 2024; Edwards and Saltzman, 2014; Saltzman et al., 2014; Saltzman and Edwards, 2017).

2.1. Great basin

The Great Basin sections span the Tremadocian to Darriwilian stages of the Lower and Middle Ordovician Series (Fig. 2). The oldest unit in this region is the House Formation, exposed at both Ibex and Shingle Pass, which is overlain by the Filmore Formation at Ibex and the Parker Spring Formation at Shingle Pass (Edwards and Saltzman, 2014). The youngest unit at Shingle Pass is the Floian-age Shingle Formation, which overlies the Parker Springs Formation. The Floian to Dapingian stages at Ibex consist of the Wah-Wah, Juab, and Kanosh formations, from base to top. The youngest units in this region are the Lehman Formation at Ibex and the Antelope Valley Formation exposed in the Antelope Range and at Meiklejohn Peak. It is worth noting that the interval we sampled at the Antelope Range and Meiklejohn Peak sections represent stratigraphic intervals with the least well-constrained biostratigraphy. For instance, in the Antelope Range section, the *Pygodus serra* – *Pygodus anserinus* biozone boundary is inferred to be within a siliciclastic unit (Member A of the Copenhagen Formation) but does not preserve any conodont materials (Harris et al., 1979). Hence, the placement of these two sections in the composite stratigraphic plots contains larger chronostratigraphic uncertainty relative to the other sections.

Previous studies on the sedimentology and facies associations of these units suggest deposition in a carbonate ramp setting. The carbonate sedimentation is thought to occur under shallow subtidal conditions with intermittent influence by tropical storms (e.g., Ross et al., 1989), as this region was located near the Palaeoequator during the Ordovician (Fig. 2). The Kanosh Formation is interpreted to have originated in a low-energy, restricted lagoonal setting (Edwards and Saltzman, 2014), which favored deposition of fine-grained sediment.

2.2. Appalachian basin

Samples collected from the Appalachian Basin sections cover the upper portion of the Darriwilian Stage and Sandbian Stage of the Middle and Upper Ordovician Series (Fig. 2). The oldest unit sampled in this study is the Darriwilian-aged Row Park Formation, exposed at the Clear Spring, Maryland section. The Row Park is overlain by the New Market Formation, exposed at both the Clear Spring and Germany Valley sections. The New Market Formation at Germany Valley is overlain by the Lincolnshire Formation. The youngest unit sampled for this study is the Chambersburg Formation exposed at Clear Spring.

Deposition during the Darriwilian to Sandbian stages in the Appalachian Basin represent a foreland basin setting. Sedimentologic studies in this region interpret the depositional environment for the Row Park, New Market, and Lincolnshire formations as shallow peritidal environments indicated by the presence of sedimentologic structures such as fenestrae in both Row Park and New Market, whereas the Chambersburg is thought to have been deposited in a more open marine subtidal setting based on the increase in invertebrate faunas (Brezinski et al., 2012). Previous chemostratigraphic studies suggest these units were deposited during a tectonic uplift event and temporary oceanic restriction in the southern Laurentian margin known as the Taconic Orogeny (e.g., Saltzman and Edwards, 2017).

3. Methods

3.1. Sample preparation and elemental analyses

Carbonate rocks were analyzed for Li isotope ratios ($\delta^7\text{Li}$) using the method outlined by (Cao et al., 2023). Samples were cut, polished, and drilled from carbonate samples, avoiding obvious veins, siliciclastic lithoclasts, or other impurities, to produce ~200 mg carbonate powders. Powdered samples were leached using ammonium acetate for 2×30 min, sonicated, and centrifuged at 5000 rpm for 10 mins. The supernatant was then filtered by passing through cellulose acetate syringe filters (pore size: 0.22 μm) to remove particles in leachates. The

supernatant was transferred to clean Teflon vials, evaporated to dryness, and re-dissolved in 0.2 N HCl for ion chromatography. A portion of the leachate was analyzed for elemental concentrations to determine elemental ratios including Li/Ca, Al/Ca, Mg/Ca, Fe/Ca, Mn/Ca, Mn/Sr, Rb/Sr, Sr/Ca, and Rb/Ca to monitor potential silicate contamination, degree of dolomitization, and diagenetic alteration (see supplement).

Elemental analyses were done at the Plasma Mass Spectrometry Laboratory at the University of North Carolina at Chapel Hill (PMS Laboratory) on an Agilent Q-ICP-MS. Aliquots for elemental concentration analyses were dissolved in 2% HNO_3 . Elemental concentrations were determined using a series of calibration standards of known concentrations. In-house standard solutions were used to correct for instrumental drift and yield an analytical precision of <2% (RSD).

3.2. Ion chromatography and isotope spectrometry

Lithium separations were done following the two-step chromatographic procedures outlined in Liu and Li (2019) to maximize recovery and minimize fractionation during lithium separation. Samples were dissolved in 0.2 N distilled HCl and loaded onto cation-exchange columns filled with Bio-Rad AG50W-X8 resin (200–400 mesh). Column recovery was monitored by collecting the pre- and post-collection cuts.

Lithium isotopic measurements were performed using an Agilent Q-ICP-MS at the PMS Lab at University of North Carolina at Chapel Hill. Samples were dissolved in 2% HNO_3 and “matrix-matched” to a concentration of 0.5 ng/g. Matrix effects were minimized by limiting the ratio of concentrations of all matrix elements to lithium by <2 (Li et al., 2020). Measurements were done in a standard – sample – standard bracketing scheme using IRMM-016 as the reference standard. In this study, we present $\delta^7\text{Li}$ values relative to the NIST I-SVEC standard as both NIST I-SVEC and IRMM-016 have practically the same $\delta^7\text{Li}$ values (Qi et al., 1997). The errors reported in this study are based on the long-term precision (reproducibility) of the measured NIST-1d standard ($2\sigma = 1.3\text{‰}$) (Cao et al., 2023).

4. Results

4.1. Lithium concentrations and elemental ratios

Lithium concentration and elemental ratio data are included in the supplementary materials (Table S2). Carbonate samples have relatively low lithium concentrations with a maximum concentration of 1.5 ng/g and an average of 0.3 ng/g. Low lithium concentrations in carbonate samples are expected because carbonate rocks are not a major Li sink in the ocean. It is worth noting that the lithium concentrations measured in this study are lower than previously published Paleozoic-aged samples (e.g., Kalderon-Asael et al., 2021), which may be due to the more rigorous leaching and filtering steps employed in this study. Due to potential variabilities associated with the methods employed in this study, diagenetic screening (i.e., statistical and covariation analyses) was based on elemental molar ratios and not [Li] alone.

Measured elemental ratios are shown in Fig. 3 and Table S2. All ratios are expressed in mmol/mol, except for Li/Ca ratios that are expressed in $\mu\text{mol/mol}$. Li/Ca ratios (proxy for Li abundance) range from 0.4 $\mu\text{mol/mol}$ to 36.9 $\mu\text{mol/mol}$, with an average of 5.9 $\mu\text{mol/mol}$. Whereas these values are lower than reported Li/Ca ratios from modern carbonate environments (e.g., the Bahamas), ratios fall within the range of previously published Paleozoic-aged carbonate rocks (e.g., Pogge von Strandmann et al., 2017). In general, $\delta^7\text{Li}$ values measured in this study do not covary with Li/Ca ratios (Pearson's $r = 0.40$). Nonetheless, some samples with anomalously high or low $\delta^7\text{Li}$ values tend to have elevated Li/Ca ratios, which may indicate contamination from a Li-rich material. Other elemental ratios do not exhibit covariation with $\delta^7\text{Li}$ values (Fig. 3). Some samples with elevated Mg/Ca and Sr/Ca ratios tend to have $\delta^7\text{Li}$ values slightly higher than the average. This may indicate variabilities in the mineralogy of samples (i.e., dolomitized carbonate

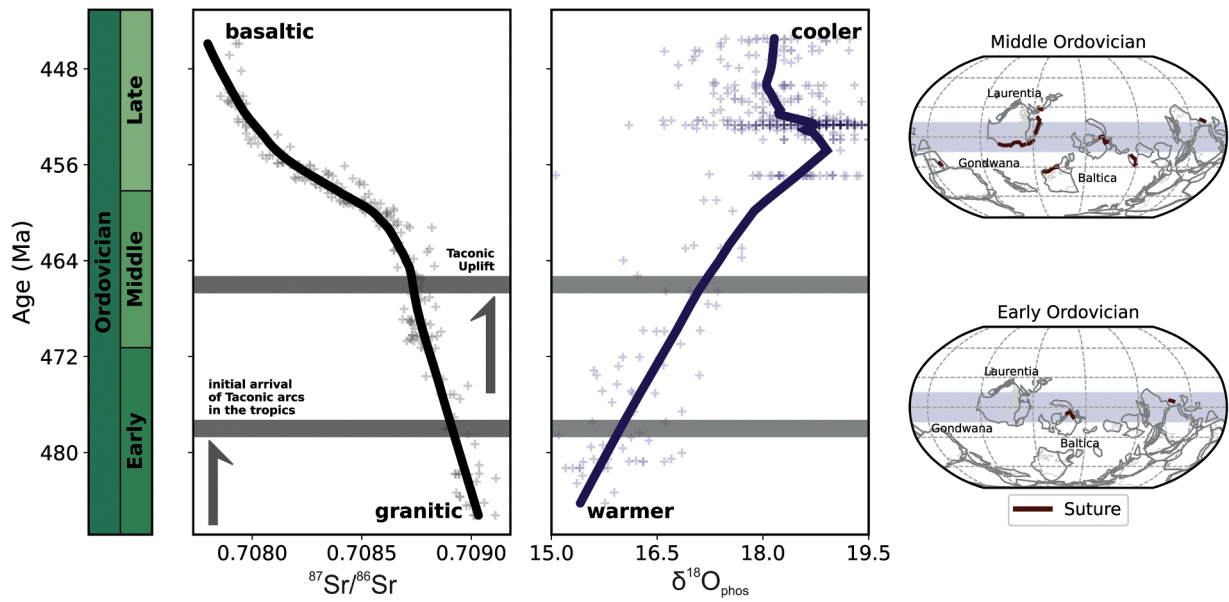


Fig. 1. Ordovician weathering, paleotemperature, and paleogeographic reconstructions. Strontium ($^{87}\text{Sr}/^{86}\text{Sr}$) data are taken from conodont measurements compiled by Avila et al. (2022) and suggest that weathering during the Ordovician is characterized by a shift from granitic to more basaltic dominant sources. The compiled conodont-based $\delta^{18}\text{O}$ data (Grossman and Joachimski, 2020) show a cooling trend throughout the Early Ordovician to the earliest Late Ordovician. Smoothing curves are constructed using LOESS whereas paleogeographic reconstructions are done using the pyGPlates Python package with a rotation file from Macdonald et al. (2019). Note that the selected LOESS bandwidth is too wide to capture a distinct cooling trend during the Middle Ordovician reported in previous studies (e.g., Avila et al., 2022; Conwell et al., 2022; Edwards et al., 2022). Red lines on the paleogeography maps represent tectonic suture zones. Age model is calibrated to GTS 2020 (Goldman et al., 2020). The timing of initial arrival and subsequent uplift of the Taconic Arcs is from Swanson-Hysell and Macdonald (2017).

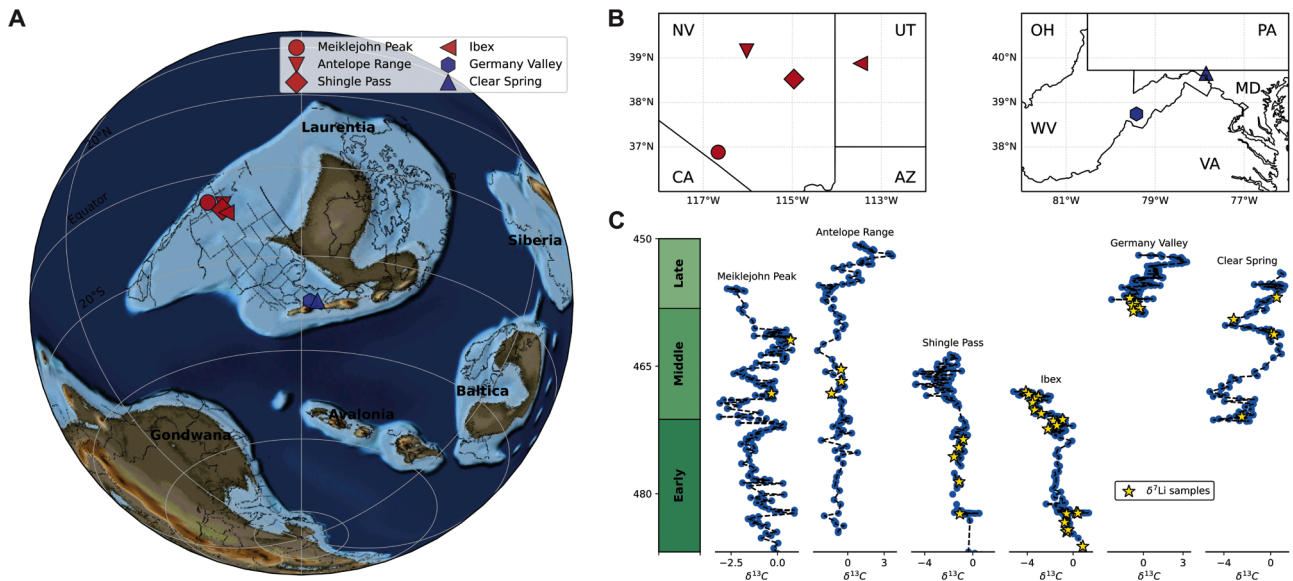


Fig. 2. Paleogeography (A), locality map (B), and carbon isotope profile (C) of the rocks used in this study. (A) The paleogeography of Laurentia and paleo-locations of the stratigraphic sections in this study. Paleogeographic base map is based on the paleoDEMS project (Scotese, and Wright, 2018) and paleo location reconstruction was done via the GPlates web service, using the rotation file of Merdith et al. (2021) and plotted with an adjustment to accommodate the discrepancy between the paleoDEMS map and the reconstructed paleo locations. (B) Present day locations of the studied sections, using the same symbology used in A. (C) Carbon isotope profiles of the studied sections taken from previously published studies (Adiatma et al., 2019; Diamond et al., 2024; Edwards and Saltzman, 2014).

and aragonite) and their associated Li fractionation factor from seawater (see section 5.1 for further discussion).

4.2. Lithium isotopic composition ($\delta^7\text{Li}$)

Results of $\delta^7\text{Li}$ measurements are listed in Table S2. The measured lithium isotopic ratios ($\delta^7\text{Li}_{\text{carb}}$) range between -0.5% to 27.7% and show a consistent temporal trend across different localities. We

reconstructed the Ordovician seawater $\delta^7\text{Li}$ ($\delta^7\text{Li}_{\text{sw}}$) by removing samples that are considered altered according to our geochemical criteria (see section 5.1 for further discussion) and adding the calcite fractionation factor ($\Delta_{\text{seawater} - \text{Calcite}}$) of 5.5% to the measured $\delta^7\text{Li}_{\text{carb}}$. The calcite fractionation factor used in this study is slightly larger than the fractionation factors used in some lithium isotope studies (e.g., Pogge von Strandmann et al., 2013). Following Cao et al. (2022), we choose $\Delta_{\text{seawater} - \text{Calcite}}$ value of 5.5% to account for possible variations in

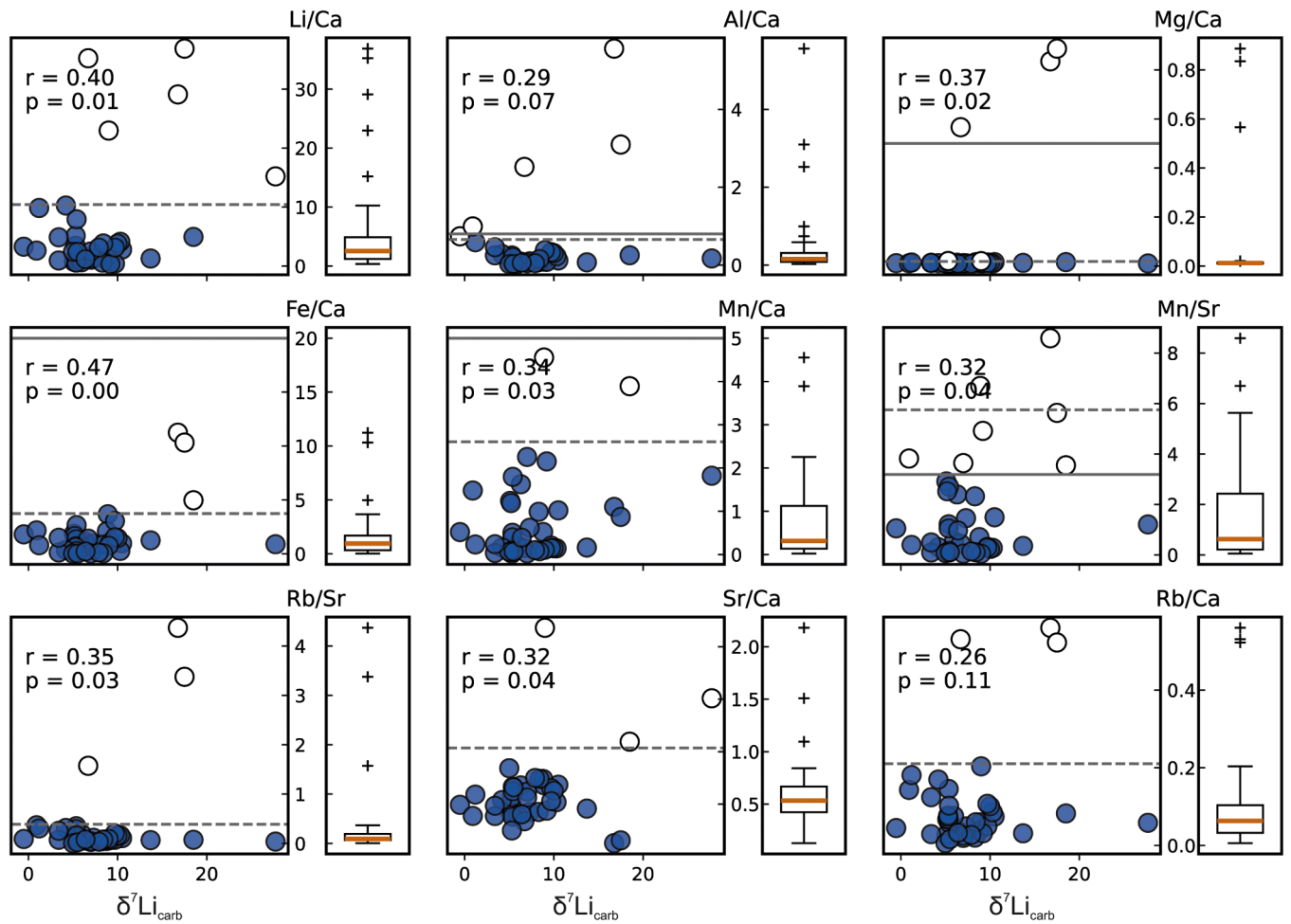


Fig. 3. Cross plots of $\delta^7\text{Li}$ vs elemental ratios and box plots used to filter potentially diagenetically altered samples in this study. Solid lines represent geochemical thresholds from previous studies, whereas dashed lines show threshold values used to define outliers (i.e., $3 \times$ interquartile range). Covariation is assessed using Pearson's r and p values shown in each cross-plot panel. Open circles in scatter plot panels show samples that are considered altered using each geochemical criterion. Cross symbols in box plot panels show outliers based on an interquartile range method.

fractionation factors associated with mineralogical variations and skeletal fragments (i.e., vital effects) in bulk carbonates (Chen et al., 2023; Dellinger et al., 2020; Washington et al., 2020). It is worth noting that the difference in the chosen fractionation factor compared to other studies is smaller than the stratigraphic variation in Ordovician $\delta^7\text{Li}_{\text{sw}}$ values reported in this study. We employ a bootstrapped LOESS method to produce a robust Ordovician $\delta^7\text{Li}_{\text{sw}}$ curve. The bootstrapping procedure was done by iteratively taking a subset of samples from the least altered $\delta^7\text{Li}$ values 500 times and generating a LOESS curve in each iteration. The final $\delta^7\text{Li}_{\text{sw}}$ curve is constructed by taking the average value of the LOESS curves from all iterations.

5. Discussion

The construction of a high-resolution $\delta^7\text{Li}_{\text{sw}}$ curve that spans the Early to early Late Ordovician period potentially enables one to reconstruct changes in the global Ordovician Li cycle, which may reflect changes in the silicon (Si) cycle, global silicate weathering patterns, and climate. Here, we first discuss and evaluate the fidelity of measured $\delta^7\text{Li}$ values, stratigraphic variabilities of $\delta^7\text{Li}$ observed in this study, and the origins of the observed variabilities in the context of changes in the Si cycle, climate, silicate weathering, and tectonic re-organization during the Ordovician.

5.1. Evaluation of diagenesis and $\delta^7\text{Li}_{\text{sw}}$ preservation

A major issue in using carbonate rocks to reconstruct changes in ocean chemistry is that shallow-water carbonates are prone to early marine diagenetic alteration that can overprint primary seawater signals (Crockford et al., 2021; Higgins et al., 2018; Braun et al., 2024; Adiatma et al., 2024). It has been suggested that the effects of diagenetic alteration depend on multiple factors, including: the diagenetic fluid composition, the reactivity of the primary mineral, the mechanism of solute transport, temperature, and pore pressure (Ahm et al., 2018). Additionally, since siliciclastic material contains several orders of magnitude more Li than carbonate rocks, the presence of siliciclastic impurities in bulk carbonate samples can play a significant role in altering $\delta^7\text{Li}_{\text{sw}}$ values. Constraining potential noise due to siliciclastic impurities is especially important in this study as some of the sections used for $\delta^7\text{Li}$ analyses are located in proximity of Taconic arcs that can be a significant source of siliciclastic input during the Ordovician. Hence, studies in $\delta^7\text{Li}$ employ strict geochemical criteria such as Al/Ca and Rb/Ca ratios to screen samples for potential siliciclastic contamination (e.g., Cao et al., 2022). In this study we opt to use a conservative approach to filter potentially altered samples by utilizing both geochemical threshold values and a statistical approach. We use the statistical method interquartile range (IQR) to filter out outliers in our samples which may indicate altered $\delta^7\text{Li}$ values. The exclusion of samples based on IQR results in lower r values (Table S3) and increased p

values, which suggests an overall weaker covariation between filtered $\delta^7\text{Li}_{\text{carb}}$, and geochemical screening criteria used in this study. A total of four samples (out of 40) were excluded based on this approach.

Variations in the mineralogy of carbonate rocks can also influence $\delta^7\text{Li}$ values. Calcite and aragonite, although composed of the same chemical composition (CaCO_3), exhibit differing $\delta^7\text{Li}$ fractionation factors during precipitation from seawater (Dellinger et al., 2020). Aragonite has a greater Li fractionation factor ($\Delta^7\text{Li}_{\text{Seawater-Aragonite}} \sim 12\text{‰}$) compared to calcite ($\Delta^7\text{Li}_{\text{Seawater-Calcite}} \sim 5.5\text{‰}$) (Dellinger et al., 2020; Marriott et al., 2004). Hence, variabilities in the primary mineralogy alone can affect the stratigraphic trend of $\delta^7\text{Li}$, particularly for studies where the change in $\delta^7\text{Li}$ values is $<6.5\text{‰}$. In this study, we use Sr/Ca ratios to constrain the effect of mineralogy in our samples, as those which were originally aragonite will typically have higher Sr/Ca ratios than calcite (Edwards et al., 2015). In general, our measured $\delta^7\text{Li}$ data do not exhibit covariation with Sr/Ca values (Pearson's $r = 0.32$), which does not appear to indicate a significant mineralogical effect in our measured $\delta^7\text{Li}$. Furthermore, the range of Sr/Ca ratios measured in this study are lower than typical modern aragonitic carbonate rocks (e.g., from the Bahamas), which may suggest that the primary mineralogy of our Ordovician samples is predominantly calcite, or the Sr/Ca ratios were lowered by post-depositional meteoric diagenesis processes (which minimally affect $\delta^7\text{Li}$ values, see Dellinger et al. (2020) for further discussion). Nevertheless, we found that the heaviest $\delta^7\text{Li}$ value measured in this study is associated with elevated Sr/Ca values. Therefore, to justify the usage of a single fractionation factor to convert $\delta^7\text{Li}_{\text{carb}}$ to $\delta^7\text{Li}_{\text{sw}}$, we exclude samples with elevated Sr/Ca values based on the IQR

threshold (Fig. 3). Additionally, we use a similar approach to exclude dolomitic samples using Mg/Ca ratios (threshold = 0.02 mol/mol).

Lastly, we use Mn/Sr ratios (mol/mol) to constrain post-depositional fluid-rock interactions that may have altered the measured $\delta^7\text{Li}$ values. As dissolved Mn is essentially absent under oxic seawater conditions, which is largely true for most of the Phanerozoic, any significant amount of Mn in carbonate rocks is likely from reducing fluid interactions during carbonate diagenesis. Most of our samples have low Mn/Sr ratios (range 0.1 – 8.6 mol/mol, mean = 1.6 mol/mol) and exhibit a lack of covariation with measured $\delta^7\text{Li}$ values. Previous studies have raised concerns over the reliability of Mn/Sr ratios in samples from ancient geological periods (i.e., Precambrian and early Paleozoic) due to frequent anoxic bottom water conditions (e.g., Kalderon-Asael et al., 2021). However, the low Mn/Sr values are consistent with recent studies using paleoredox proxies suggest that the carbonate rocks used in this study were not formed in anoxic or euxinic conditions (e.g., Lu et al., 2023). Therefore, we adopt a conservative approach and apply a geochemical threshold of 3 mol/mol (following Kaufman et al. (1993)) to exclude the possibility of diagenetically altered samples (Fig. 3). Additionally, calcium isotopes ($\delta^{44/40}\text{Ca}$) and strontium isotopes ($^{87}\text{Sr}/^{86}\text{Sr}$), in combination with screening criteria utilized here, may provide a way to further test the hypothesis of primary seawater $\delta^7\text{Li}$ values (Cao et al., 2022; Crockford et al., 2021) but this could only be done for a small number of samples analyzed here (Adiatma et al., 2024) and a statistically meaningful analysis is not yet possible.

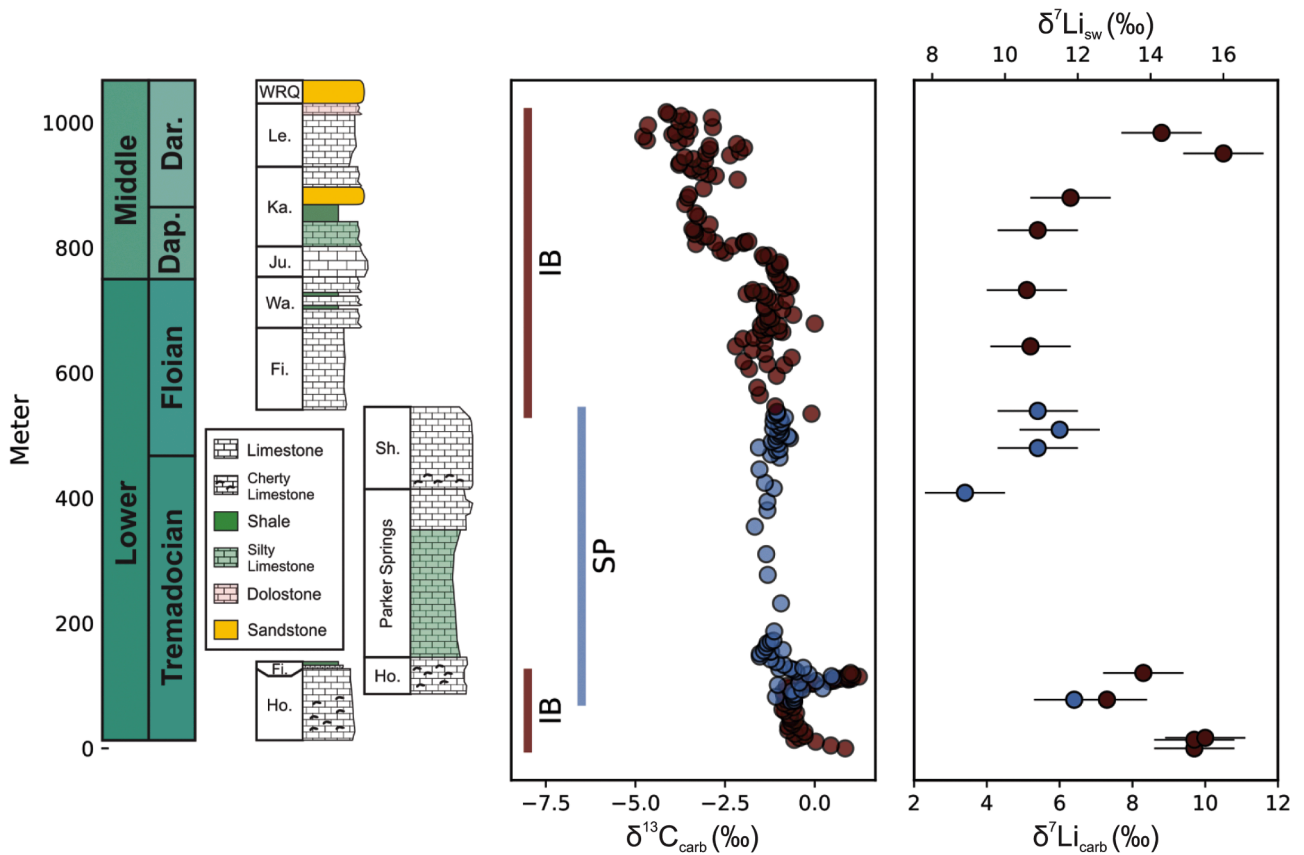


Fig. 4. Composite lithostratigraphic and isotope chemostratigraphic profiles from sections in the eastern Great Basin region. Stratigraphic sections shown in the composite profiles are Ibex (IB), Shingle Pass (SP). Lithium isotope data ($\delta^7\text{Li}$, ‰ I-SVEC) are from screened samples. Samples from the Parker Springs Formation are not used in this study due to their high clay content. Conversion to $\delta^7\text{Li}_{\text{sw}}$ was done by assuming calcite fractionation factor of 5.5‰. Carbon isotope ($\delta^{13}\text{C}$, ‰ V-PDB) data are taken from Edwards and Saltzman (2014). Error bars shown in the $\delta^7\text{Li}$ plot are based on long-term reproducibility ($2\sigma = 1.3\text{‰}$) of NIST 1d standards (Cao et al., 2022). Formation abbreviations: Ho. = House; Fi. = Fillmore; Wa. = Wah; Ju. = Juab; Ka. = Kanosh; Le. = Lehman; WRQ. = Watson Ranch Quartzite; Sh. = Shingle Limestone.

5.2. Stratigraphic variations in Ordovician seawater $\delta^7\text{Li}$

Reconstructed Ordovician $\delta^7\text{Li}_{\text{sw}}$ values show a long-term ($> 10^7$ years) baseline average between $\sim 11\text{‰}$ and 15‰ that is significantly less than younger periods in Earth history (Kalderon-Asael et al., 2021). On time scales of millions of years, comparable to the residence time of Li in the oceans (Misra and Froelich, 2012), our Ordovician dataset appears to show small sample-to-sample variation ($< 2\text{‰}$) around this long-term average (Fig. 4). Furthermore, this trend includes a gradual $\sim 4\text{‰}$ negative shift that characterizes the Lower Ordovician interval of the studied sections. Statistical analyses done on the $\delta^7\text{Li}$ dataset generated in this study suggest that the observed gradual shift is statistically significant (see supplementary materials for further discussion). Additionally, the magnitude of the shift observed in this study is comparable to previously reported $\delta^7\text{Li}$ variations in other geologic time periods that have been interpreted to be associated with LIP emplacement (e.g., $\sim 5\text{‰}$ negative shift associated with the Deccan Traps (Misra and Froelich, 2012), hyperthermal events (e.g., $\sim 4\text{‰}$ negative shift associated with the Paleocene Eocene Thermal Maximum event (Pogge Von Strandmann et al., 2021), and climate cooling (e.g., ~ 8 to 10‰ positive shift associated with the Hirnantian Glaciation (Pogge von Strandmann et al., 2017).

5.3. The Lower Ordovician negative $\delta^7\text{Li}$ shift: a link to silicate weathering?

The Lower Ordovician interval of the studied sections is characterized by a gradual decrease in $\delta^7\text{Li}_{\text{sw}}$ values from $\sim 15\text{‰}$ to $\sim 11\text{‰}$ (Fig. 4). The consistency across different sections that span this interval supports the notion that the observed decline in the measured $\delta^7\text{Li}_{\text{carb}}$ may reflect a global trend (Fig. 5). If the declining trend in $\delta^7\text{Li}$ values represents a decline in $\delta^7\text{Li}_{\text{sw}}$, there are three possible main drivers of this trend: enhanced reverse weathering, enhanced seafloor basaltic alteration, and a shift toward a congruent weathering regime.

Reverse weathering or the formation of marine authigenic clay minerals could have potentially driven Early Ordovician $\delta^7\text{Li}_{\text{sw}}$ to lower values. During formation of authigenic clay minerals, ^6Li is preferentially incorporated into clay minerals which should lead to heavier $\delta^7\text{Li}_{\text{sw}}$ values (Tomascak et al., 2016). However, a rapid uptake of Li

during an episode of enhanced reverse weathering can lead to a near complete removal of Li from seawater that could result in very limited isotopic fractionation between seawater and clay minerals (Kalderon-Asael et al., 2021). In other words, as the dissolved Li reservoir becomes progressively smaller during intense reverse weathering, the fractionation factor of $\delta^7\text{Li}$ associated with the reverse weathering process ($\Delta^7\text{Li}_{\text{sed}}$) approaches zero (i.e., preferential uptake of ^6Li during clay formation becomes weaker) and lower the total $\Delta^7\text{Li}_{\text{sink}}$. Therefore, a combination of progressive enhanced reverse weathering and limited fractionation between seawater and clay minerals could produce a negative baseline shift in $\delta^7\text{Li}_{\text{sw}}$. A study by Cao et al. (2022) for instance, argues that the observed rapid decline of $\delta^7\text{Li}_{\text{sw}}$ at the end of the Permian is likely due to this mechanism. However, unlike the end of the Permian that is thought to be characterized by a weakening of the biogenic silica pump (as shown by the lack of chert deposition, known as the “Permian Chert Gap”), which led to an increase in reverse weathering rates, the Cambrian to Ordovician period is characterized by a radiation of marine silicifying organisms such as sponges and radiolarians (Maliva et al., 2005; Servais et al., 2016). The increase in diversity of silicifying organisms should increase the biogenic silica pump, which in turn weakens global reverse weathering rates due to the presence of the additional Si sink. This notion is supported by sedimentologic evidence of the presence of chert or cherty limestone across the Ordovician study interval (e.g., the House Formation) (Edwards and Saltzman, 2014). Therefore, it is unlikely that the observed decline in $\delta^7\text{Li}_{\text{sw}}$ during the Early Ordovician was caused by changes in reverse weathering rates alone.

The second possible cause of a decline in $\delta^7\text{Li}_{\text{sw}}$ is enhanced low-temperature alteration of oceanic crust (AOC) coupled with a decrease in the fractionation factor. One possible driver of enhanced AOC is increased seafloor production (Li and West, 2014). A gradual decline in $^{87}\text{Sr}/^{86}\text{Sr}$ that spans the Early to Middle Ordovician has been suggested as a potential indication of an increase in seafloor production during this interval (Avila et al., 2022). However, alteration of oceanic crust preferentially incorporates ^6Li and leaves seawater enriched in ^7Li . Consequently, a transient change in AOC alone (in a relatively short time scale) should drive $\delta^7\text{Li}$ values to be more positive, unless it is accompanied by a lowering in the fractionation factor (Δ_{AOC}). One way to lower Δ_{AOC} is by increasing the temperature at which the water – rock

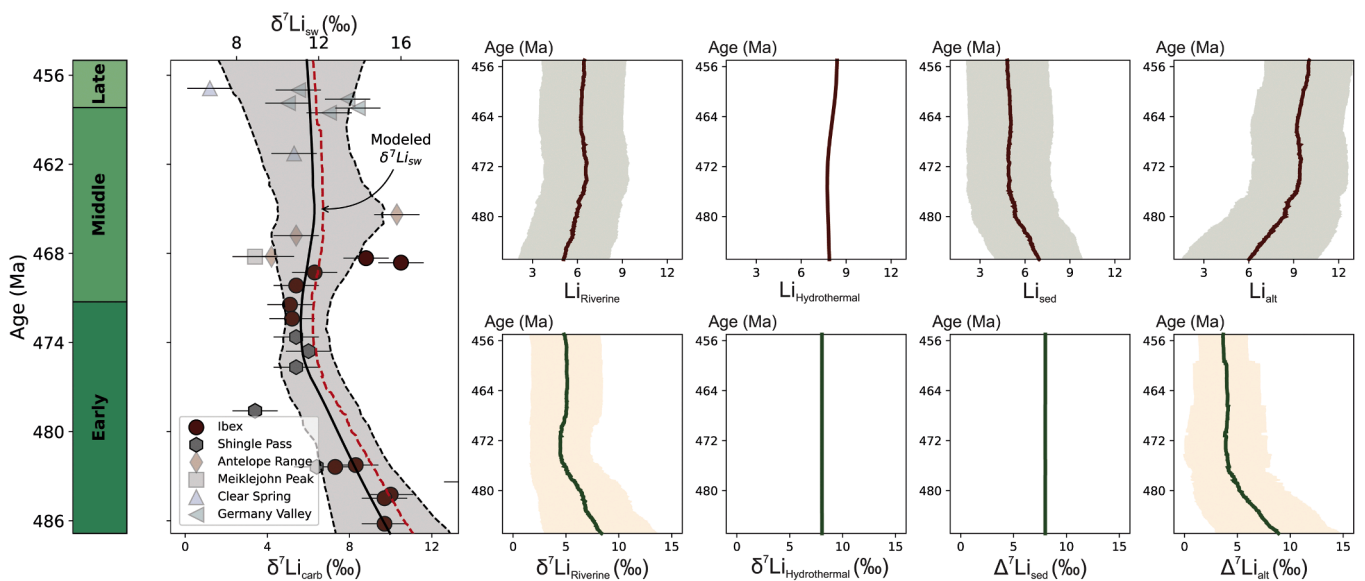


Fig. 5. Results of the stochastic Li mass balance model. The solution space is shown as lines and error bands. The lines and error bands represent the mean and standard deviations of solutions in each timestep respectively. Lithium fluxes are shown in Gmol/yr (10^9 mol/yr). The riverine flux solution space shows a 31% increase during the Early Ordovician. The modeled $\delta^7\text{Li}_{\text{sw}}$ is shown as a red dashed line. Comparison using root mean squared error method (RMSE) between data vs modeled $\delta^7\text{Li}_{\text{sw}}$ and data vs LOESS curve are 2.25 and 2.20 respectively.

interaction takes place, as lithium isotope fractionation during water-rock interaction is a temperature-dependent process (lower fractionation at higher temperatures) (Marschall et al., 2007). Furthermore, a previous study suggested that ocean bottom water temperatures are likely to be a controlling factor in any temperature-dependent processes within the seafloor hydrothermal system, including AOC (Coogan and Dosso, 2015). This implies that a lowering in Δ_{AOC} requires an increase in the ocean bottom water temperature. While some studies have suggested a possible warming episode during this portion of the Ordovician (e.g., Edwards et al., 2022), the overall $\delta^{18}O$ trend during the Early to Middle Ordovician reflects cooling period (Goldberg et al., 2021; Grossman and Joachimski, 2020; Trotter et al., 2008). Therefore, it is unlikely that this mechanism drove the decline in $\delta^{7}Li_{sw}$ during the Ordovician.

Here we propose that the most likely driver for this Early Ordovician negative $\delta^{7}Li_{sw}$ shift is a protracted change in the global weathering regime toward more congruent weathering (low-intensity weathering regime *sensu* Pogge von Strandmann et al., 2021) due to the migration of an island arc into the tropics (Swanson-Hysell and Macdonald, 2017). This interpretation is supported by several lines of evidence. First, the decline in seawater $\delta^{7}Li$ during the Early Ordovician is accompanied by a decrease in $^{87}Sr/^{86}Sr$ (Fig. 1), which may imply a common driver. Seawater $^{87}Sr/^{86}Sr$ can be affected by both changes in the hydrothermal flux and continental weathering. If an increase in seafloor production had been the driver of changes in both the Li and Sr cycles during the Early Ordovician, the process would result in an increase in the Li hydrothermal flux to the ocean and potentially enhanced Li sink due to basaltic alteration in off-axis hydrothermal systems (e.g., enhanced hydrologic cycle associated with elevated seafloor topography (Coogan et al., 2017)), which together shifted seawater $\delta^{7}Li$ toward lower values. For instance, if we assume a source and sink proportion that resembles the modern Li cycle (Li and West, 2014), a 50% increase in hydrothermal Li flux and AOC sink will translate into $\sim 2\text{‰}$ negative $\delta^{7}Li$ shift. This suggests that inducing the observed shift in seawater $\delta^{7}Li$ during the Ordovician would require an increase in seafloor production by $>50\%$ which is unlikely to be the case during the Early Ordovician. Moreover, a global plate reconstruction suggests that the Early Ordovician is characterized by a decline in seafloor production rates (Merdith et al., 2021), implying that changes in continental weathering are the most likely driver of changes in both the Sr and Li cycles during this period.

A protracted change in the global weathering regime toward more congruent weathering during the Early Ordovician is also supported by the evidence for the northward migration of an island arc (the Taconic Arc) toward the tropics. While classical interpretations of Laurentian paleogeography place the arrival of the Taconic Arc in the tropics during the Late Ordovician, a more recent paleogeographic reconstruction, utilizing new paleomagnetic data, indicates an earlier arrival in the tropics during the Early Ordovician (Swanson-Hysell and Macdonald, 2017). The northward migration of an island arc is a likely driver of changes in the weathering regime, as this type of geologic province is characterized by easily weatherable mafic rocks that may be exposed in steep slope topography. It is worth mentioning that constraining the paleo-topography of the Taconic Arc is not trivial as this region has undergone multiple orogenic events and intense weathering. Therefore, future studies that focus on reconstructing the paleo-topographic conditions of this region are required to further test this model. Nevertheless, the combination of easily weatherable mafic rocks and steep topography could potentially lead to a more congruent weathering regime (Bouchez et al., 2013). A congruent weathering regime is characterized by a high riverine flux of lithium with a low $\delta^{7}Li$ isotopic composition. It is worth noting that the most prominent signal of Taconic Arc weathering is documented within the Middle Ordovician Series (Conwell et al., 2022; Swanson-Hysell and Macdonald, 2017). Additionally, it is notable that modern island arcs are characterized by thick regolith, indicating an incongruent weathering regime, which is

inconsistent with our proposed model. However, we argue that modern day island arcs may not be a perfect analogue for Ordovician island arcs (i.e., the Taconic Arc), as modern island arcs are dominated by deep-rooted (vascular) land plants that enhance soil formation which did not exist during the Ordovician Period (Morris et al., 2018). The lack of deep-rooted land plants during the Ordovician may be a significant factor that contributed to the shift toward congruent weathering rather than incongruent weathering, which is the dominant weathering regime in modern island arcs.

A complementary or possibly alternative explanation for a shift toward a more congruent weathering regime is an increase in the overall exposed landmass during this period. A recent paleogeographic reconstruction by Marcilly et al. (2021) shows that the Early Ordovician is characterized by the onset of global increase in exposed landmass within the tropics. A protracted change in weathering regime, therefore, can alter the isotopic composition of seawater toward lower $\delta^{7}Li$ values as observed in our Early Ordovician dataset.

We further explore the potential cause of the observed negative shift in our $\delta^{7}Li$ data by constructing a stochastic inverse lithium box model. This approach can reduce bias in model parameterization and explore the full range of possible forcings that can cause the variability in our observations. Specifically, we use a modified isotopic mass balance equation (Eq. (1) and Eq. (2)) from Li and West (2014), where F_r , F_h , F_{sed} , and F_{alt} are Li sources and sinks associated with continental weathering (riverine flux: F_r), hydrothermal input (F_h), reverse weathering (F_{sed}), and alteration (F_{alt}) of oceanic crust. R_r and R_h are the isotopic composition of the continental weathering flux and hydrothermal flux, respectively. The Δ_{sed} and Δ_{alt} correspond to lithium fractionation factors associated with reverse weathering and alteration of oceanic crust respectively.

$$\delta^7Li_{sw} = \frac{(F_r \times R_r) + (F_h \times R_h) + (F_{sed} \times \Delta_{sed}) + (F_{alt} \times \Delta_{alt})}{F_{sed} + F_{alt}} \quad (1)$$

$$F_r + F_h = F_{sed} - F_{alt} \quad (2)$$

The model relies on a Monte Carlo resampling method to explore a wider range of possible solution space of forcings (i.e., Li fluxes, ratios, fractionation factors) that match the $\delta^{7}Li_{sw}$ curve constructed in this study. The parameter resampling and calculations were performed 1000 times in each time-step. The produced arrays of solutions were then filtered using the Ordovician $\delta^{7}Li_{sw}$ curve with a tolerance band of 1.3‰ that reflects the analytical uncertainty of our method.

Fig. 5 shows the results of the stochastic inverse model with the additional constraints of the hydrothermal flux based on the seafloor production curve of Merdith et al. (2021). This also includes a constant fractionation factor for reverse weathering (Δ_{sed}) and the $\delta^{7}Li$ of hydrothermal flux (R_h), which reflects our current understanding of the oceanic Si cycle. Additionally, we vary Δ_{alt} from 1‰ to 30‰ . This relatively large range of values is based on the range of values associated with temperature dependent Li fractionation during rock-fluid interaction (Li and West, 2014). The results of this modeling exercise show that the observed Ordovician $\delta^{7}Li_{sw}$ trend can be produced by a decrease in $\delta^{7}Li_{riv}$ coupled with an increase in the continental weathering flux. This implies a progressive change toward a low intensity congruent weathering regime during the Early Ordovician as a likely driver for the negative baseline shift. These results support the hypothesized increasing silicate weathering rates during the Ordovician due to enhanced global weatherability associated with the arrival of Taconic arc in the tropics and overall increase in exposed landmass.

Besides a shift in both the riverine flux and $\delta^{7}Li_{riv}$, the stochastic model outputs also show changes in Li sinks associated with both the reverse weathering flux and AOC (Fig. 5). While a decline in reverse weathering is consistent with increasing diversity of silicifying organisms during the Great Ordovician Biodiversification Event (Maliva et al., 2005), there is no evidence for an elevated Li flux associated with AOC. In hypothetical scenarios simulated using a deterministic reverse

modeling scheme (see supplementary materials) where we hold both sinks constant, the required $\delta^7\text{Li}_{\text{riv}}$ to reproduce the observed baseline shift during the Ordovician is very negative, lower than the average upper continental crust (e.g., Tomascak et al., 2016). The results of these hypothetical scenarios suggest that the observed negative shift in seawater $\delta^7\text{Li}$ during the Ordovician requires either changes in Li sinks (reverse weathering and/or AOC) or that the increase in the riverine flux suggested by the stochastic model represents a minimum value and the overall increase to induce the observed baseline shift in seawater $\delta^7\text{Li}$ was higher. The latter also implies that during the Early Ordovician the oceanic Li cycle may not be in a steady state condition.

Previous studies also suggest that terrestrialization by non-vascular land plant during the Ordovician can be a potential forcing that enhanced silicate weathering and permanently changed the global carbon cycle (e.g., Adiatma et al., 2019; Lenton et al., 2016; Pogge von Strandmann et al., 2017; Sproson et al., 2022). Non-vascular land plants (e.g., lichens and liverworts) lack rooting structures that can induce pedogenesis and clay formation. Therefore, enhanced silicate weathering by non-vascular land plants could have led to a congruent weathering regime (Pogge von Strandmann et al., 2017; Sproson et al., 2022). However, the precise timing of non-vascular land plant colonization, which is largely based on a molecular clock technique and cryptospore fossils, contains large uncertainties (e.g., Lenton et al., 2016; Morris et al., 2018). Recent molecular clock studies suggest that early terrestrialization by non-vascular land plants may have taken place as early as the Cambrian (Morris et al., 2018; Servais et al., 2019). This finding raises the possibility that terrestrialization by non-vascular land plants may have contributed to a shift toward a more congruent weathering regime during the Ordovician. Nevertheless, future studies to further constrain the timing of this event may provide a test of this hypothesis.

It is notable that the Middle Ordovician portion of the curve is characterized by a plateau in the $\delta^7\text{Li}$ values as this interval is thought to have recorded a transient increase in basaltic weathering due to the uplift of the Taconic island arc (Conwell et al., 2022; Swanson-Hysell and Macdonald, 2017). The combined effect of steeper topography and

increased supply of basaltic rocks should have driven riverine $\delta^7\text{Li}$ toward lighter values. The plateauing of $\delta^7\text{Li}_{\text{sw}}$ is likely due to a possible increase in the hydrothermal Li flux, which balances out the increased continental Li flux with lighter $\delta^7\text{Li}$ values.

5.4. The Ordovician $\delta^7\text{Li}$ baseline and Si cycles

Our estimated $\delta^7\text{Li}_{\text{sw}}$ values are between 11‰ and 15‰ and fall within the range of values previously reported for Paleozoic seawater (Fig. 6) (Kalderon-Asael et al., 2021) and are significantly lower than younger periods in Earth history that reach values as high as 31‰ in the modern (Misra and Froelich, 2012). A previous study suggested that the lower $\delta^7\text{Li}$ values observed in the Precambrian and early Paleozoic interval are due to a weaker isotope fractionation during Li burial coupled with rapid clay formation rates (Kalderon-Asael et al., 2021). Further, Kalderon-Asael et al. (2021) argue that the overall changes toward heavier $\delta^7\text{Li}_{\text{sw}}$ during the Phanerozoic reflects a progressive decrease in marine Si concentrations, which may be linked to a shift toward a more biologically-controlled Si cycle due to the emergence of silicifying marine organisms and proliferation of land plants (Kalderon-Asael et al., 2021; Morris et al., 2018). The emergence of land plants has been thought to have shifted clay production from the ocean to land, which eventually weaken the global clay production via reverse weathering (Kalderon-Asael et al., 2021). It is worth noting, however, that the $\delta^7\text{Li}_{\text{sw}}$ curve produced for the Early to Middle Ordovician period in this study exhibits a trend opposite to the long-term trend reported by Kalderon-Asael et al. (2021), despite the emergence of non-vascular land plants (Morris et al., 2018) and an increase in diversity of silicifying organisms during the Great Ordovician Biodiversification Event (GOBE) (Maliva et al., 2005; Servais et al., 2016). This suggests that changes in the Li cycle during the Early to Middle Ordovician may have been decoupled from the proposed long-term changes in the marine Si cycle and are likely to be controlled by changes in the continental Li input.

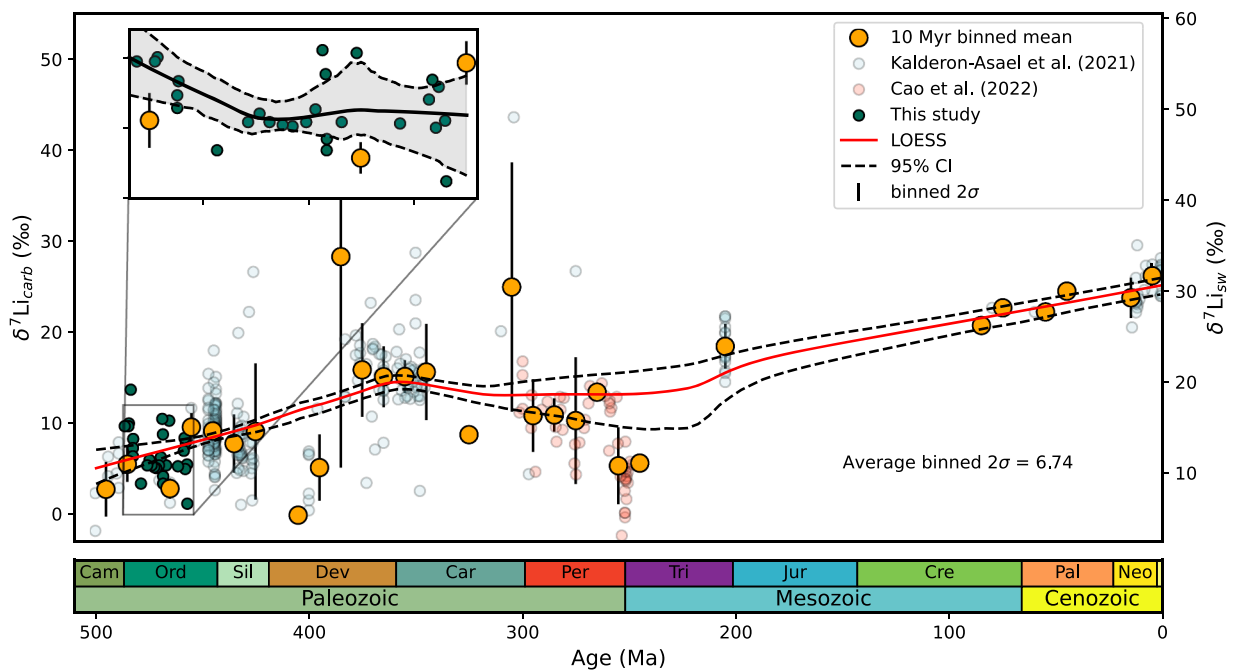


Fig. 6. Phanerozoic $\delta^7\text{Li}_{\text{carb}}$ compilation with a conversion to $\delta^7\text{Li}_{\text{sw}}$. Compilation based on data from Kalderon-Asael et al. (2021), Cao et al. (2022), and this study. Large circles represent 10 Myr average of the Phanerozoic $\delta^7\text{Li}$ compilation (Cao et al., 2023; Kalderon-Asael et al., 2021). LOESS smoothing (red line) was done on data from Kalderon-Asael et al. (2021) and Cao et al. (2023). The LOESS curve shown in the inset only includes data from this study. Conversion to $\delta^7\text{Li}_{\text{sw}}$ was done by assuming calcite fractionation factor of 5.5‰.

6. Conclusions

Using a suite of carbonate rock samples from six sections across the Laurentian paleocontinent, combined with a rigorous textural and geochemical screening, this study provides the first high-resolution $\delta^{7}\text{Li}_{\text{sw}}$ curve that spans the Early to early Late Ordovician period. The $\delta^{7}\text{Li}_{\text{sw}}$ curve exhibits a 4‰ negative baseline shift during the Early to Middle Ordovician. Based on available geologic data and numerical modeling employed in this study, a shift toward a low intensity congruent weathering regime may reflect an increase in the availability of weatherable rocks in the tropics and possible terrestrialization by non-vascular land plants. This finding suggests that the global Li cycles during the Ordovician were mainly controlled by changes in silicate weathering instead of Si cycles. Furthermore, the shift toward more congruent weathering may have enhanced the efficiency of CO_2 consumption via silicate weathering and likely contributed to the Early Ordovician cooling trend.

CRediT authorship contribution statement

Y. Datu Adiatma: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Conceptualization. **Matthew R. Saltzman:** Writing – review & editing, Supervision, Resources, Conceptualization. **Xiao-Ming Liu:** Writing – review & editing, Resources, Methodology. **Xi-Kai Wang:** Writing – review & editing, Methodology. **Cole T. Edwards:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The python code and data related to this article is available online at <https://github.com/datuadiatma/stochastic-lithium>.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.epsl.2024.119030](https://doi.org/10.1016/j.epsl.2024.119030).

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